

NRCM HACKATHON · PNT1 · LEVEL 1

AEGIS Twin

Tech stack, roadmap & risk register

Phases 2–11 · Day 1 = 1 Oct 2026

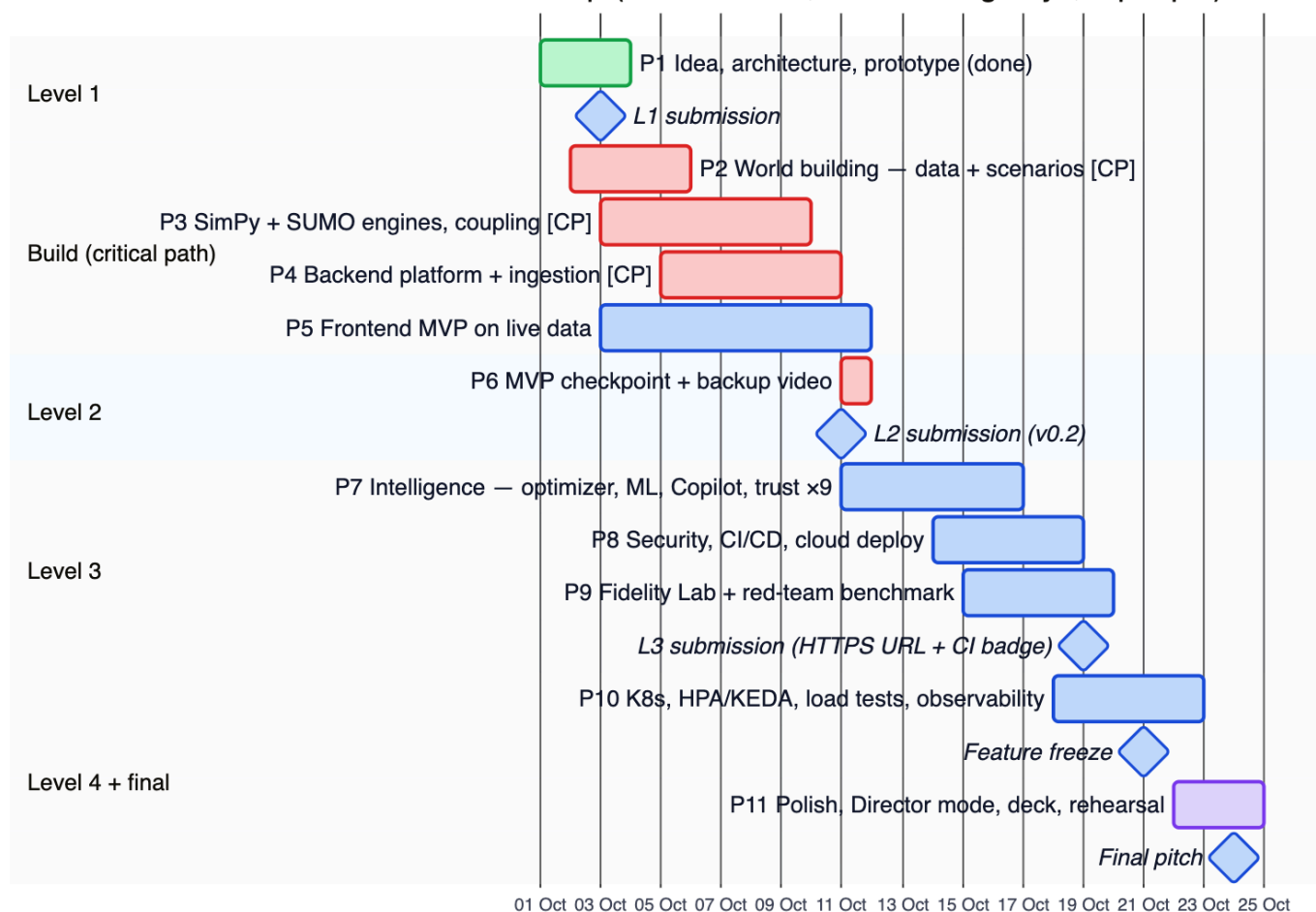
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Level 1 deliverable · 30 September 2026

Dating assumption: Day 1 = 1 Oct 2026, with ~4 people and ~24 working days. Once the organisers confirm the level deadlines, shift the bars; the order and dependencies stay the same.

AEGIS Twin — build roadmap (Phases 2–11, ~24 working days, 4 people)



Roadmap Gantt, Phases 2–11

1. Phase plan

Owners: **P1** simulation lead · **P2** platform lead · **P3** frontend lead · **P4** intelligence + story lead. [CP] = critical path.

Phase	Dates (Day)	Owner	Scope	Exit criterion
1 Idea, architecture, planning	1–3 Oct (D1–3)	all	SRS, C4, sequences, DFDs, ERD, clickable prototype, roadmap, research slide	✅ This submission
2 World building [CP]	2–5 Oct (D2–5)	P1	<code>data/nodes.json</code> , <code>lanes.json</code> , <code>skus.json</code> (done early in Phase 1); demand model; SUMO Hyderabad net; 7 scenario-DSL templates	<code>python -m sim.macro.run --days 30</code> prints KPIs; <code>python -m sim.micro.run</code> moves trucks
3 Engines + coupling [CP]	3–9 Oct (D3–9)	P1	SimPy macro engine, MC runner, SUMO runner, coupling orchestrator, Reality Emulator + attack injector	Coupled twin runs live; emulator streams; engine verification tests pass (SRS §8); NFR-3 benchmarked
4 Backend + ingestion [CP]	5–10 Oct (D5–10)	P2	Alembic schema (§7), ingest API, Redis Streams, twin-state service, WS fan-out, REST, scenario jobs + spec-hash cache	Emulator → ingest → twin → WS pipeline visible in <code>wscat</code> ; first NFR-1 / NFR-2 numbers
5 Frontend MVP	3–11 Oct (D3–11)	P3	Swap the prototype's mock JSON for live WS/REST; client interpolation; Scenario Lab v1 on real results	Control Tower on live data at 60 fps
6 Level 2 checkpoint	11 Oct (D11)	all	End-to-end: live map → Chennai closure → MC → results → impact on map; tag <code>v0.2</code> ; backup video; README	Level 2 submission
7 Intelligence	11–16 Oct (D11–16)	P4, P2	Optimizer (reroute, min-cost-flow, expedite, buffer), Pareto/CVaR ranking, cascades + REI, LightGBM ETA, demand forecast, Copilot, full 9-layer trust	Ranked plans + Apply move trucks; each attack type caught by its layer
8 Security + CI/CD + cloud	14–18 Oct (D14–18)	P2	JWT + RBAC, HMAC devices, Caddy TLS, rate limits, audit log, GitHub Actions (lint/type/test/Playwright/Trivy), GHCR, VM deploy, Vercel	Public HTTPS URL with login; green CI badge
9 Fidelity Lab	15–19 Oct (D15–19)	P4	Shadow predictions, live metrics, drift recalibration, red-team benchmark (≥ 500 attacks), counterfactual decision value, DataCo backtest, <code>/eval/report</code>	Measured headline numbers for the pitch
10 Scale + reliability (Level 4)	18–22 Oct (D18–22)	P2	Per-service containers, Helm, HPA + KEDA, Prometheus/Grafana/Loki/Alertmanager, k6/Locust, circuit breakers, backups, chaos drill	Helm chart; Grafana screenshots; load-test report; pod kill self-heals live
11 Polish + pitch	continuous; 22–24 Oct (D22–24)	P3, P4	Director mode, backup video, 10-slide deck, Q&A prep, 5+ rehearsals on the projector	Feature freeze D21; final pitch D24

Dependencies: P3 needs the P2 network data. P4 needs P3's `Twin.snapshot()` API. P7 optimisation needs the P3 Monte Carlo runner. P9 needs P3's Reality Emulator (ground truth) and P7's trust layer. P10 needs the P8 containers.

2. Tech stack

Layer	Choice
Simulation	SimPy 4.1 (macro, real-time env), Eclipse SUMO 1.27 via libsumo (micro), NetworkX
Optimisation / ML	OR-Tools (min-cost flow, VRP), PuLP + HiGHS, LightGBM, statsforecast, scikit-learn, filterpy
AI	Claude API with tool use (<code>claude-sonnet-5-5</code> routing, <code>claude-opus-5-5</code> reasoning), evidence-gated
Backend	FastAPI, Pydantic v2 strict, uvicorn, orjson, msgpack; Redis 7 (Streams, pub/sub, cache); PostgreSQL 16 + PostGIS + TimescaleDB
Frontend	Vite, React 19, TypeScript, Tailwind v4, deck.gl 9.4, MapLibre 5, ECharts 6, Motion, Zustand, cmdk, sonner, react-force-graph-3d
DevOps	GitHub Actions, Docker, Docker Compose, k3s + Helm, HPA/KEDA, Caddy, Prometheus, Grafana, Loki, Alertmanager, k6, Locust, Vercel

3. Risk register

Likelihood (L) and impact (I): H/M/L. Each risk has an owner, an early-warning trigger and a mitigation already scheduled in the plan.

ID	Risk	L	I	Owner	Trigger (early warning)	Mitigation	Contingency
R-1	SUMO performance: network too big or slow to run faster than real time with the target vehicle count	M	H	P1	D3 benchmark: 500 trucks + 3,000 cars slower than 1× real time	Small bbox (western + southern belt, not the whole ORR); <code>--keep-edges.by-type</code> motorway→tertiary; <code>--geometry.remove --junctions.join</code> ; libsumo in-process (not TraCI sockets); cap live trucks at 500; low-density <code>randomTrips</code> background; SUMO in its own process	Pre-record SUMO FCD for the demo corridor and replay it; Monte Carlo is already macro-only (ADR-0003)
R-2	Time sync: macro and micro clocks drift, so shipments arrive twice, never, or out of order	M	H	P1	Handshake test flaky; arrivals outside ±1 step	A single orchestrator owns the clock ; SUMO is stepped to the macro <code>RealtimeEnvironment</code> (e.g. 1 sim-min per wall-second); a command queue with sim-time stamps; CI integration test (spawn in macro → arrive in micro → received in macro); monotonic IDs make receipts idempotent	Fall back to macro-only live mode with SUMO as visual replay

ID	Risk	L	I	Owner	Trigger (early warning)	Mitigation	Contingency
R-3	Demo failure: Wi-Fi, cloud, tile CDN or LLM down on stage	M	H	P3	Any rehearsal needing a network retry	Director mode (F) replays the whole story locally from a recorded seed; backup 1080p video (local + unlisted YouTube); pre-cached map tiles; cached Copilot answers for the scripted question; phone hotspot; laptop runs the full docker compose stack offline	Switch to the video at the first sign of trouble, and narrate live
R-4	Scope creep	H	M	P4	Any MUST item behind by > 1 day	MoSCoW cut list (<code>PLAN.md §8</code>); feature freeze D21	Cut from COULD, then SHOULD; never MUST
R-5	Judges doubt the data is real	M	M	P4	Q&A rehearsal	Say it upfront: Reality Emulator (ADR-0002) + real OSM, weather, AIS, DataCo; measured numbers only	Show the DataCo backtest slide
R-6	LLM latency or cost during the demo	M	M	P4	p95 Copilot latency > 4 s	Cache scripted-question responses; fast model for tool routing	Director mode plays the cached transcript
R-7	Projector washes out the dark UI	M	M	P3	First projector test	Test at projector resolution early; high-contrast theme toggle; bump the ink colour	Light theme
R-8	Ingest misses NFR-1 on 4 vCPU	M	M	P2	First k6 run < 3,000 msg/s	Batching, orjson, XADD pipelining, uvicorn workers = vCPU, trust workers scaled by KEDA	Report the measured ceiling honestly, with the profile
R-9	Team availability (exams, illness)	M	M	all	Missed stand-up	Pairing across roles; docs in-repo; a Director-mode demo is always runnable	Re-plan via the cut list

4. Definition of done (every phase)

Code merged via PR with CI green. Tests for new logic. The demo path still runs in Director mode. Numbers quoted anywhere are reproducible from a seed.